Day 9 Report: What People Say About LLMs on Twitter

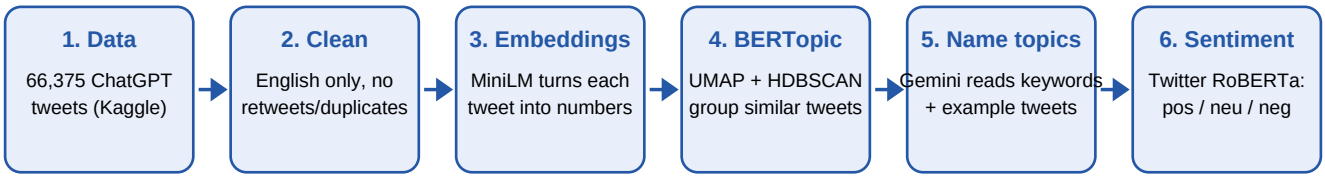
Topic modelling (BERTopic) + topic naming with an LLM + sentiment analysis (Twitter RoBERTa)

1. Goal

Find out **what** people tweet about ChatGPT and other LLMs, **how they feel** about it, how this **changes over time**, and whether **popular accounts** talk differently from average users.

2. How it works

Pipeline



3. Data

Step	Tweets left
Raw dataset	66,375
English only	20,545
Without retweets	7,389
Without duplicates (spam bots)	6,981
Without very short tweets (final)	6,673

Date range: 3 April 2023 to 12 May 2023 (about 108 to 291 tweets per day).

4. Results

Topics. BERTopic found **16 topics**. 1,181 tweets (18%) did not fit any topic (outliers). The biggest topic by far is *ChatGPT Prompting and Usage* (3,265 tweets), followed by *Future of Artificial Intelligence* (743).

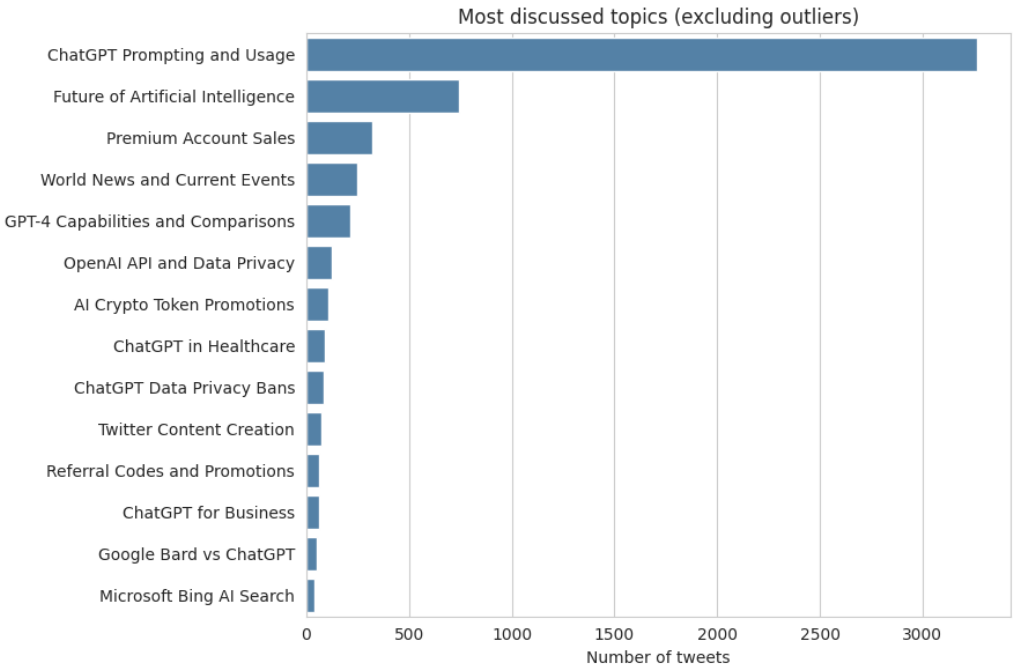


Figure 1. Topic sizes (topics named by the LLM)

Sentiment. Overall: **49.6% neutral, 31.9% positive, 18.6% negative**.

Topic	Negative %	Positive %	What it means
ChatGPT Data Privacy Bans	27.6	6.9	Most negative: worries about bans and data
OpenAI API and Data Privacy	25.4	14.8	Privacy concerns
ChatGPT Prompting and Usage	24.3	26.3	Mixed: people share wins and frustrations
Future of Artificial Intelligence	17.9	33.5	More hopeful than worried
AI Crypto Token Promotions	0.0	99.1	Spam / adverts, always "positive"
Referral Codes and Promotions	0.0	98.5	Spam / adverts

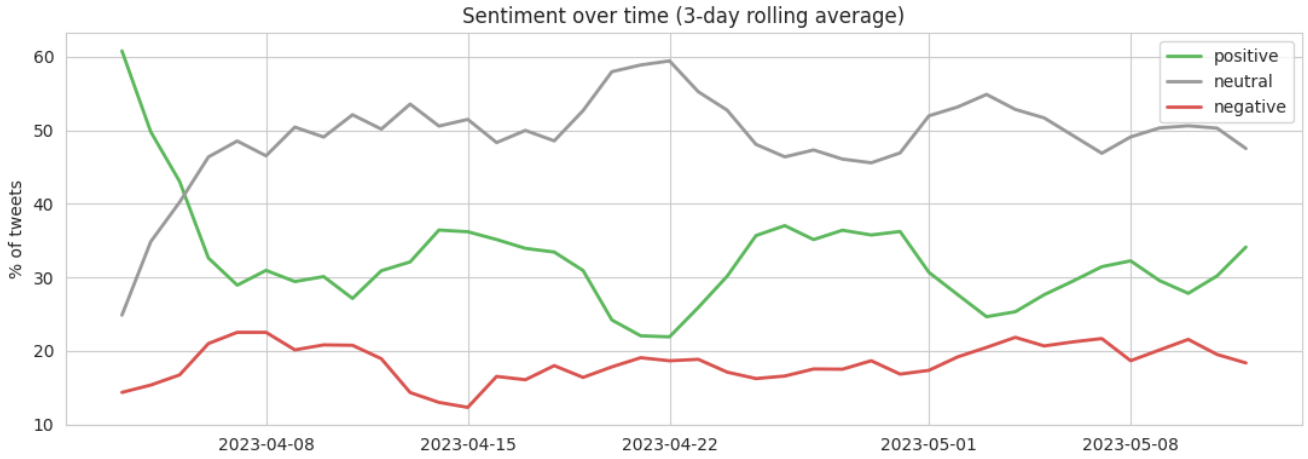


Figure 2. Sentiment over time (3-day rolling average)

Over time. Neutral tweets are always the largest group (about 45-60%). Positive sentiment dipped around 20-22 April and recovered later. Negative stays fairly steady at about 15-22%.

Popular vs average accounts. Popular = top 10% by followers (5,137+ followers, 668 tweets). Their sentiment is almost the same as average users (18.0% vs 18.6% negative), just slightly more neutral. Their tweets get more views (median 89.5 impressions vs 27).

Engagement. Negative tweets get the most attention on average: 17.8 likes and 2,192 impressions, compared with about 7-8 likes for neutral and positive tweets.

5. Key findings

- Most LLM talk is practical: how to use and prompt ChatGPT.
- Privacy and bans are the most negative topics.
- Topics that look "very positive" (crypto, referral codes) are mostly **spam**, so a high positive score does not always mean real happiness.
- Negative tweets spread more (more likes and views), which is common on social media.
- Popular accounts and average users feel about the same; popular accounts just get more reach.

6. Problems and fixes

- Three "Premium Account Sales" topics got the same name: the LLM gave similar names to similar spam groups. These could be merged.
- CUDA error at the start: fixed by letting the code choose the device automatically.
- Colab restarts lost variables: fixed by saving results to CSV and using Run all.

7. Limitations and next steps

- Only about 6 weeks of tweets, and only English.
- The sentiment model was used as-is; a small hand-labelled check would show how accurate it is.
- Next: remove spam before topic modelling and merge duplicate topics.